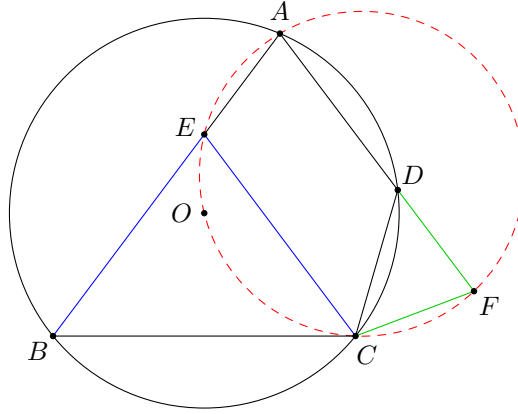




HK IMO Training 2022-2023
Problem Set 1 Suggested Solutions



Problem 1 (Benelux MO 2021 Problem 3). Let $ABCD$ be a cyclic quadrilateral with circumcentre O . Let E be a point on the line AB such that $EB = EC$. Let F be a point on the line AD such that $FD = FC$. Prove that the circumcentre of $\triangle AEF$ lies on the perpendicular bisector of CO .



Solution. We use directed angles. Since O is the centre of (ABC) , we have

$$\angle AOC = 2\angle ABC = \angle EBC + \angle BCE = \angle AEC.$$

This implies A, E, O, C are concyclic. Similarly, we have

$$\angle AOC = 2\angle ADC = \angle FDC + \angle DCF = \angle AFC.$$

This implies A, O, C, F are concyclic. Therefore, A, E, O, C, F are concyclic. It is now obvious that the circumcentre of $\triangle AEF$ lies on the perpendicular bisector of CO . $\square$

Problem 2 (Balkan MO 2017 Problem 1). Solve the equation $m^3 + n^3 = m^2 + 42mn + n^2$ in positive integers.

Solution. Let $d = (m, n)$ and let $m = da, n = db$ where $(a, b) = 1$. The equation becomes

$$d(a^3 + b^3) = a^2 + 42ab + b^2. \quad (1)$$

Note that $a^2 - ab + b^2$ divides $a^3 + b^3$, hence $a^2 - ab + b^2 \mid a^2 + 42ab + b^2$. This implies $a^2 - ab + b^2 \mid 43ab$. As $(a, b) = 1$, we deduce $(a^2 - ab + b^2, a) = (b^2, a) = 1$ and similarly $(a^2 - ab + b^2, b) = 1$. Therefore, we must have $a^2 - ab + b^2 \mid 43$. This yields $a^2 - ab + b^2 = 1, 43$ since $a^2 - ab + b^2 = \left(a - \frac{b}{2}\right)^2 + \frac{3b^2}{4}$ is positive.

- If $a^2 - ab + b^2 = 1$, then $(2a - b)^2 + 3b^2 = 4$. As b is a positive integer, we must have $b = 1$ and $2a - b = \pm 1$. The only possibility is $(a, b) = (1, 1)$ since $a > 0$. Equation (1) becomes $2d = 44$, hence $d = 22$ and $m = n = 22$. This is a solution since the given equation is equivalent to (1).
- If $a^2 - ab + b^2 = 43$, then $(2a - b)^2 + 3b^2 = 172$. Clearly, $b \leq 7$ from $3b^2 \leq 172$. When $b = 1, 2, 3, 4, 5, 6, 7$, we have $(2a - b)^2 = 169, 160, 145, 124, 97, 64, 25$ respectively. This leads to the only possible positive integer solutions $(a, b) = (7, 1), (7, 6), (1, 7), (6, 7)$. When $(a, b) = (7, 1), (1, 7)$, equation (1) becomes $344d = 344$, hence $d = 1$ and $(m, n) = (7, 1), (1, 7)$. Again, both are solutions. When $(a, b) = (7, 6), (6, 7)$, equation (1) becomes $559d = 1849$. Then $d = \frac{43}{13}$ is not an integer.

Therefore, the only solutions are $(m, n) = (22, 22), (7, 1), (1, 7)$. $\square$

Problem 3 (Benelux 2016 Problem 3). Find all functions $f : \mathbb{R} \rightarrow \mathbb{Z}$ such that

$$f(f(y) - x)^2 + f(x)^2 + f(y)^2 = f(y) + 2f(y)f(f(y)) \quad (1)$$

for any $x, y \in \mathbb{R}$.

Solution. Putting $x = y = 0$ in (1), we obtain

$$f(f(0))^2 + 2f(0)^2 = f(0) + 2f(0)f(f(0)).$$

This implies $f(0) - f(0)^2 = [f(f(0)) - f(0)]^2 \geq 0$. As $f(0)$ is an integer, we have $f(0) = 0, 1$.

- If $f(0) = 0$, we put $y = 0$ in (1) to obtain $f(-x)^2 + f(x)^2 = 0$. Clearly, this implies $f(x) = 0$ for any $x \in \mathbb{R}$. This is a solution by simple checking.
- If $f(0) = 1$, we put $x = y = 0$ in (1) to obtain $f(1)^2 + 2 = 1 + 2f(1)$. This implies $(f(1) - 1)^2 = 0$, and hence $f(1) = 1$. Next, putting $y = 0$ in (1), we get $f(1 - x)^2 + f(x)^2 + 1 = 1 + 2$, i.e. $f(1 - x)^2 + f(x)^2 = 2$. Therefore, we must have $f(1 - x) = \pm 1$ and $f(x) = \pm 1$. Next, putting $x = 0$ in (1) gives

$$f(f(y))^2 + 1 + f(y)^2 = f(y) + 2f(y)f(f(y)).$$

This implies $f(y) - 1 = (f(f(y)) - f(y))^2 \geq 0$, hence $f(y) = 1$. It is clear that this is another solution.

The solutions are $f(x) = 0$ for all $x \in \mathbb{R}$ and $f(x) = 1$ for all $x \in \mathbb{R}$. □

Problem 4 (RMM 2018 Problem 3). Two players X and Y play a game on an infinite grid as follows. Starting from X , the two players take turns choosing an undirected edge of the grid and assign a direction to this edge (in one of the two directions parallel to this edge). Player Y wins if there exists a directed cycle in the grid. Determine if Y has a winning strategy.

Solution. No, X can always prevent Y from winning the game.

We set up the coordinate system in the obvious way. We pair up the edges $\{(x, y), (x - 1, y)\}$ and $\{(x, y), (x, y - 1)\}$ for any x and any $y \geq 1$, and we pair up the edges $\{(x, y), (x + 1, y)\}$ and $\{(x, y), (x, y + 1)\}$ for any x and any $y \leq -1$. The remaining unpaired edges all lie on the x -axis. The strategy of X is first to assign a direction to any edge on the x -axis. Afterwards, whenever Y assigns a direction to an edge in a pair, X assigns a direction to another edge in the same pair such that it is impossible to travel along these two edges consecutively. Whenever Y assigns a direction to an edge on the x -axis, X assigns a direction to any edge on the x -axis arbitrarily. We claim that this prevents the formation of a directed cycle.

Consider the vertices of any directed cycle, if exists. Let (a, b) be the vertex with the greatest y -coordinate among all vertices and the greatest x -coordinate among all vertices having y -coordinate b . Then it is clear that we have to use the edges $\{(a, b), (a - 1, b)\}$ and $\{(a, b), (a, b - 1)\}$ to pass through the point (a, b) . In view of the construction, we must have $b \leq 0$. Similarly, let (c, d) be the vertex with the smallest y -coordinate among all vertices and the smallest x -coordinate among all vertices having y -coordinate d . Then we have to use the edges $\{(c, d), (c + 1, d)\}$ and $\{(c, d), (c, d + 1)\}$ to pass through the point (c, d) . It follows that $d \geq 0$. But then we have $0 \leq d \leq b \leq 0$. This implies all vertices in this cycle lie on the x -axis, which is impossible. Therefore, Y can never win. □